

Cross-Platform Mobile App Development

Week 3 — Progressive Web Apps (PWA)

Instructor: Nguyen Thanh Tuan, PhD

Institution: Faculty of Computer Science, VKU

Today's Agenda

1. What makes a web app a *Progressive* Web App (PWA)
2. Web App Manifest (`manifest.json`) configuration & Standalone display mode
3. Service Worker Lifecycle: Install → Activate → Fetch
4. 5 core caching strategies: Cache-First, Network-First, Stale-While-Revalidate, Cache-Only, Network-Only
5. IndexedDB for local structured storage & Offline draft persistence
6. Background Sync API & offline queue synchronization
7. Mini-Project 1 Kickoff: VKU Field Survey App

Part 1: What is a Progressive Web App?

5 Criteria of a PWA

-  **Installable** — Added to home screen with standalone launcher icon
-  **Offline-First** — Operates reliably without network connectivity via Service Worker + Cache API
-  **Fast & Responsive** — Sub-second boot times from local cache, adapts to all mobile viewports
-  **Engaging** — Web Push notifications and badging API
-  **Secure** — Delivered strictly over HTTPS

PWA vs. Native App Comparison

Feature	Progressive Web App	Native Mobile App
Distribution	Direct URL / Web Link	Google Play / Apple App Store
App Store Fee	0% Fee	15% – 30% Fee
Updates	Instant (via Service Worker)	Store Review (1–3 Days)
Bundle Size	1 – 5 MB	25 – 80 MB
Hardware Access	Good (Camera, GPS, Storage)	Full (BLE, NFC background, Biometrics)
Offline Support	Cache API + IndexedDB	Local SQLite / File System

Part 2: Service Worker & Cache API

Service Worker Lifecycle

[Register in main.js]



(Install Event) —> Pre-cache core App Shell assets (HTML, CSS, JS, Icons)



(Activate Event) —> Clean up deprecated cache versions



(Fetch Event) —> Intercept HTTP requests & apply caching strategies

5 Core Caching Strategies

1. **Cache-First (App Shell):** Check Cache first; if found, return immediately; else fetch from network.
2. **Network-First (Live Data):** Attempt fresh network request; fallback to Cache when offline.
3. **Stale-While-Revalidate (Social / Feed):** Return cached response instantly while fetching update in background.
4. **Cache-Only:** Restrict to pre-cached offline assets.
5. **Network-Only:** Direct pass-through for non-cacheable transactional requests (e.g. Auth tokens).

Part 3: Offline Data & Mini-Project 1

IndexedDB for Structured Offline Data

- Local key-value object store built into all modern mobile browsers.
- Asynchronous API handling megabytes of structured data, blobs, and inspection records.
- Works hand-in-hand with Service Workers to queue offline survey submissions.

Mini-Project 1: VKU Field Survey PWA

- **Goal:** Offline-first inspection form for campus facilities with zero network connectivity.
- **Key Deliverables:**
 - i. Live deployed HTTPS link (Cloudflare Pages / Vercel).
 - ii. Public GitHub repository with clean code and README setup.
 - iii. Short Technical Report (2–4 pages PDF).
- **Next Week:** Wrapping this PWA into an Android APK using Capacitor Bridge.